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Communication Protocol

SK-AD3 Series

Card Issuing Machine



REV: A 2011-01-01

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1 Communication Format Descriptions

1.1 Communication Character Format

Baud Rate (BPS): 9600/192000/38400/57600BPS (Automatic recognition)
Communication Method: Asynchronous
Transmission Type: Half-duplex, multi communication with max 16 C/Ds.
Data frame Structure:

Start bit	D0	D1	D2	D3	D4	D5	D6	D7	Stop sbit

Start Bit: 1 bit
Data Bit: 8 bits
Parity Bit: N/A
Stop Bit: 1 bit
Coding Scheme: ASCII 8-bit encoding

1.2 Communication Control Method and Control Character

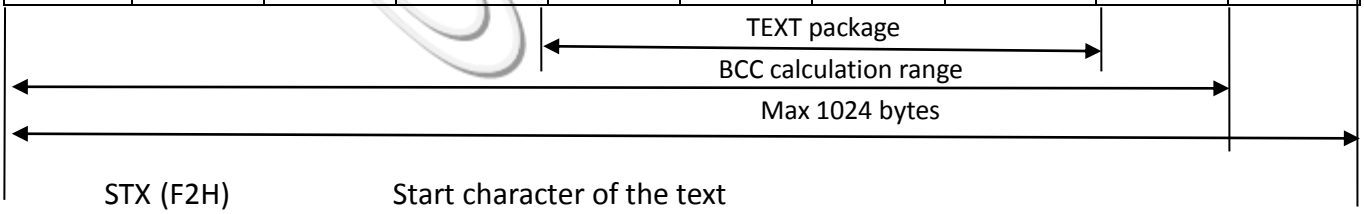
C/D is slave part which can operate only after receiving HOST effective commands.
Related control character:

ACK (06H) Acknowledge
NAK (15H) Negative acknowledge
EOT (04H) End of transmission

1.3 Communication Format and Related Character

1.3.1 Send Command Package Format

STX (0xF2)	ADDR (1 byte)	LENH (1 byte)	LENL (1 byte)	CMT (1 byte)	CM (1 byte)	PM (1 byte)	DATA (N bytes)	ETX (1 byte)	BCC (1 byte)

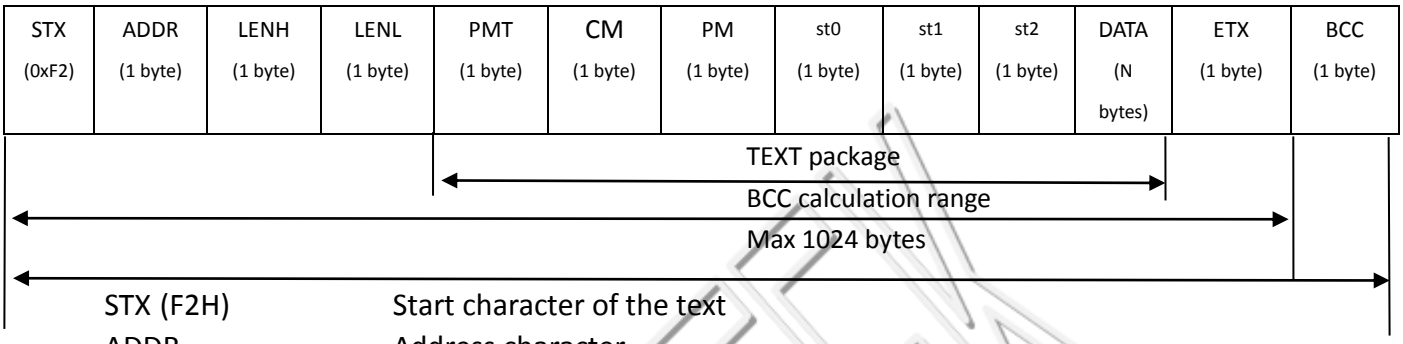


STX (F2H) Start character of the text
ADDR Address character
LENH (1 byte) High byte of the TEXT length
LENL(1 byte) Low byte of the TEXT length
CMT Command header ('C', 43H)
CM Command code
PM Parameter
DATA Send data command (N byte, N=0~512)
ETX (03H) End character of the text

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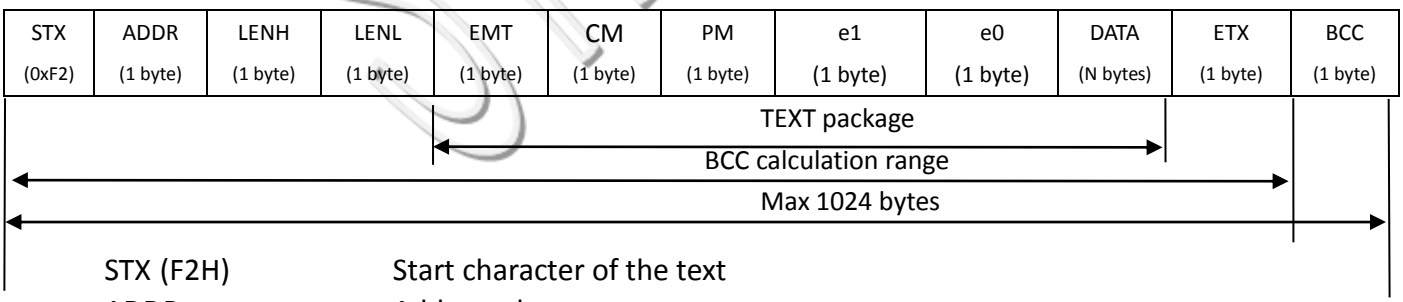
BCC (1 byte)	Block Check Character

1.3.2 Command Package Format for Success Return



STX (F2H)	Start character of the text
ADDR	Address character
LENH (1 byte)	High byte of the TEXT length
LENL (1 byte)	Low byte of the TEXT length
PMT	Command header (‘P’, 50H)
CM	Command code
PM	Parameter
st1,st0,st2	Status code
DATA	Send data command (N byte, N=0~512)
ETX (03H)	End character of the text
BCC (1 byte)	Block Check Character

1.3.3 Command Package Format for Failure Response



STX (F2H)	Start character of the text
ADDR	Address character
LENH (1 byte)	High byte of the TEXT length
LENL (1 byte)	Low byte of the TEXT length
EMT	Command header (‘N’, 45H)
CM	Command code
PM	Parameter
e1,e0	Error code
DATA	Send data command (N byte, N=0~512)
ETX (03H)	End character of the text

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BCC (1 byte)	Block Check Character

2 Multi Communication Address

ADDR: Corresponding address for multi communication is as follow:

C/D Address	ADDR
0#	00H
1#	01H
2#	02H
3#	03H
4#	04H
5#	05H
6#	06H
7#	07H
8#	08H
9#	09H
10#	0AH
11#	0BH
12#	0CH
13#	0DH
14#	0EH
15#	0FH

Default ADDR is 15# before delivery. For multi communication, please set each C/D as a unique ADDR and control each C/D by corresponding ADDR in the communication package.

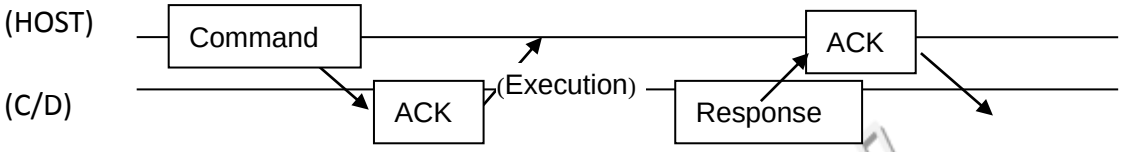
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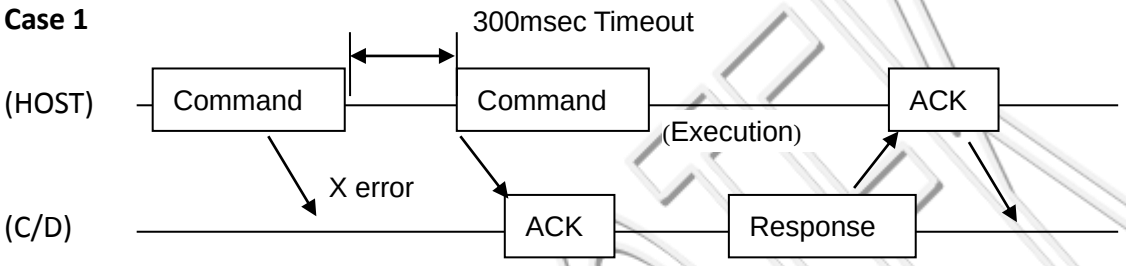
3 Communication Process Descriptions

3.1 Normal Communication Procedure (Command and Response):

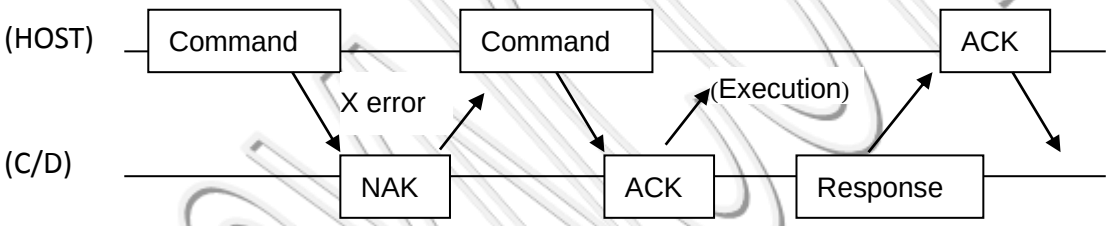


3.2 Abnormal Communication Procedure

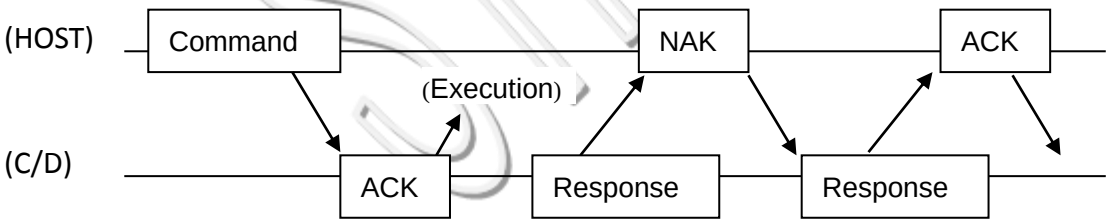
Case 1



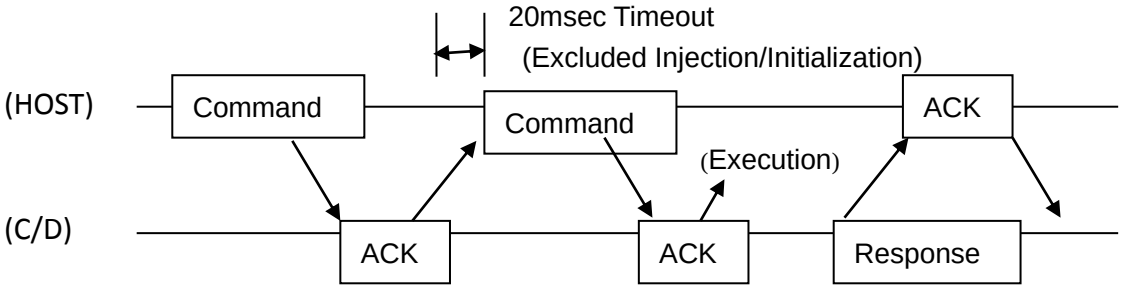
Case 2



Case 3

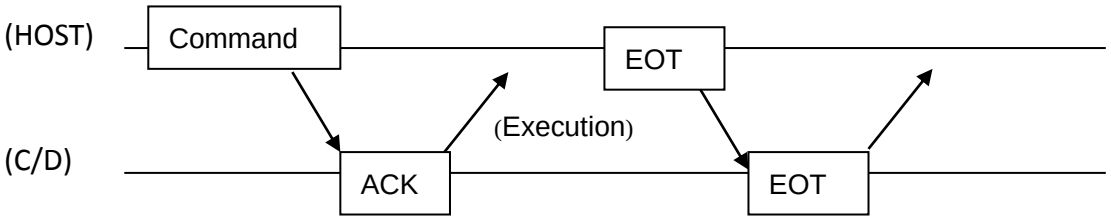


Case 4

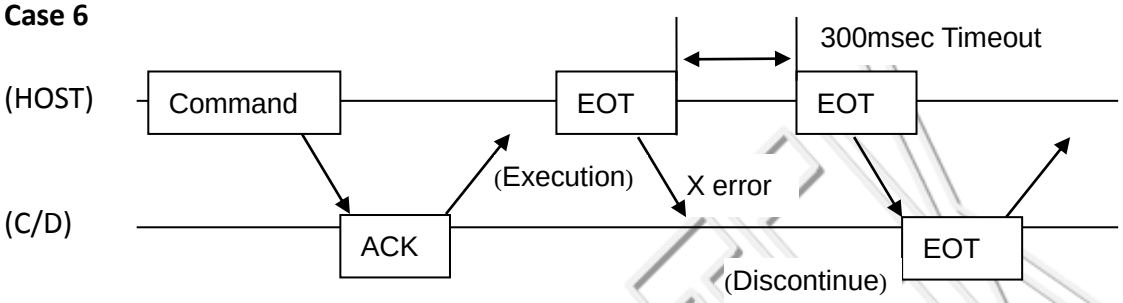


Case 5

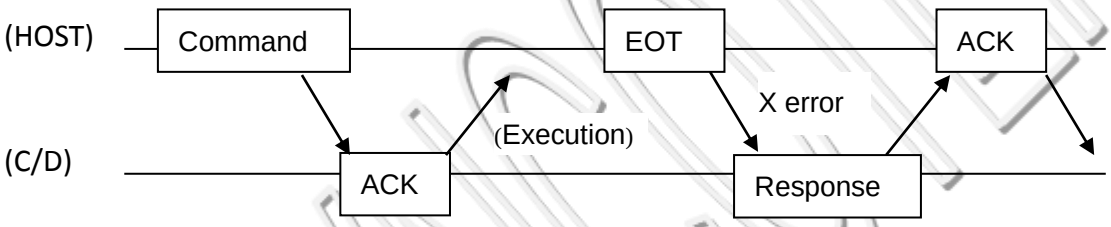
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Case 6



Case 7



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4 Status Codes and Error Codes

4.1 Status Codes

st0	DESCRIPTIONS
“0”	No card inside C/D.
“1”	One card inside C/D.
“2”	Card in RF/IC position.

st1	DESCRIPTIONS
“0”	No card inside stacker.
“1”	A few cards inside stacker.
“2”	Sufficient cards inside stacker.

st2	DESCRIPTIONS
“0”	Stacker capacity not full..
“1”	Stacker capacity full.

4.2 e1, e0 Error Codes

e1, e0	DESCRIPTIONS
“00”	Undefined command.
“01”	Command parameter error.
“02”	Command execution sequence error.
“03”	Hardware doesn’t support command.
“04”	Command data error in communication package.
“05”	IC card contact deactivates.
“06—“09”	
“10”	Card jam.
“11”	
“12”	Sensor error.
“13”	Too long card.
“14”	Too short card.
“15”—“39”	
“40”	Card is withdrawn when retracting.
“41”	IC card solenoid error.
“42”	
“43”	Disable to move to IC position.
“44”	
“45”	Card is moved by outer force.
“46”	

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“47”	
”48”	
“49”	
“50”	Counter overflow.
“51”	Motor error.
“52”—“59”	
“60”	IC card power failure.
“61”	IC card activation failure.
“62”	IC card doesn’t support current command.
“63”	
“64”	
“65”	IC card deactivates.
“66”	Current IC card doesn’t support any command.
“67”	IC card data transmission error.
“68”	IC card data transmission timeout.
“69”	CPU/SAM card doesn’t conform to EMV standard.
“A0”	Stacker empty or no card inside stacker.
“A1”	Stacker capacity full.
“A2”—“A9”	
“B0”	C/D not reset.

5 Basic Command Descriptions

5.1 Reset/Initialize

HOST command:

“C”	30H	Pm

Success returns:

“P”	30H	Pm	st0	st1	st2	Rev_type

Failure returns:

“N”	30H	Pm	e1	e0

This command should be executed first once power is on otherwise the other command cannot be executed. For first execution, C/D will auto test HOST baud rate and communicate with this BAUD. Once execution, all error codes will be deleted, C/D is under INSERTION DENIAL status and returns software version information.

Pm: C/D reset handling parameter.
If no card is inside C/D, motor will rotate slightly (to prepare the card in the stacker).
If card is inside C/D, it will handle as following:

= 30H: Moves card to front holding position.
= 31H: Moves card to capture box.
= 33H: Not moves card.
= 34H: As 30H and starts counter function.

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= 35H: As 31H and starts counter function.
= 37H: As 33H and starts counter function.

Rev_type: C/D software version information.

5.2 Status Sense

HOST command:

"C"	31H	Pm

Success returns:

"P"	31H	Pm	st0	st1	st2	Sensor (12 bytes)

Failure returns:

"N"	31H	Pm	e1	e0

Pm = 30H: Reads C/D condition, st0, st1, st2 (see 7. Status Codes for details).
Pm = 31H: Reads C/D condition and return sensors (12 bytes) status information as following:

SENSOR	STATUS
S1	30H: Card inside.
	31H: No card inside.
S2	30H: Card inside.
	31H: No card inside.
S3	30H: Card inside.
	31H: No card inside.
S4	30H: Card inside.
	31H: No card inside.
S5 (Reserved)	
S6	30H: Card inside.
	31H: No card inside.
S7	30H: Card inside.
	31H: No card inside.
S8	30H: Card inside.
	31H: No card inside.
S9	30H: Card inside.
	31H: No card inside.
S10	30H: Card inside.
	31H: No card inside.
KS1	30H: Card inside.
	31H: No card inside.
KS2	30H: Card inside.
	31H: No card inside.

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5.3 Move card

HOST command:

"C"	32H	Pm

Success returns:

"P"	32H	Pm	st0	st1	st2

Failure returns:

"N"	32H	Pm	e1	e0

Pm = 30H: Moves card to front holding position.
Pm = 31H: Moves card to IC position (only for stacker dispensing).
Pm = 32H: Moves card to RF position.
Pm = 33H: Moves card to capture box.
Pm = 39H: Moves card to gate position.

If command cannot move card to specified position, C/D will report CARD JAM error.
Notes: When executing card retraction while stacker is full, C/D will report STACKER FULL error, please clear up the stacker and card.

5.4 Set insertion

HOST command:

"C"	33H	Pm

Success returns:

"P"	33H	Pm	st0	st1	st2

Failure returns:

"N"	33H	Pm	e1	e0

Set insertion enabled/disabled. "ENABLED" function will move card to RF position once there is a card insertion from the bezel, canceling insertion could be done by STATUS SENSE.

Pm = 30H: Permits insertion.
Pm = 31H: Denies insertion.

After executing RESET/INITIALIZE, C/D will deny insertion.

5.5 Auto test IC/RF card type

5.5.1 Auto test IC card type

HOST command:

"C"	50H	30H

Success returns:

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“P”	50H	30H	st0	st1	st2	Card_type

Failure returns:

“N”	50H	30H	e1	e0

Auto test current IC card type and move card to IC position, then report card type information.

Card_type (2 bytes)		Descriptions
“0”	“0”	Unknown IC card type
“1”	“0”	T=0 CPU card
	“1”	T=1 CPU card
“2”	“0”	SLE4442 card
	“1”	SLE4428 card
“3”	“0”	AT24C01 card
	“1”	AT24C02 card
	“2”	AT24C04 card
	“3”	AT24C08 card
	“4”	AT24C16 card
	“5”	AT24C32 card
	“6”	AT24C64 card
	“7”	AT24C128 card
	“8”	AT24C256 card

5.5.2 Auto test RF card type

HOST command:

“C”	50H	31H

Success returns:

“P”	50H	31H	st0	st1	st2	Card_type

Failure returns:

“N”	50H	31H	e1	e0

Auto test current RF card type and move card to RF position, then report card type information.

Card_type (2 bytes)		Descriptions
“0”	“0”	Unknown RF card type
“1”	“0”	Mifare one S50 card
	“1”	Mifare one S70 card
	“2”	Mifare one UL card
“2”	“0”	Type A CPU card
“3”	“0”	Type B CPU card

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6 Contact IC Card Operate

6.1 Operate CPU card

6.1.1 CPU card cold reset

HOST command:

"C"	51H	30H	Vcc

Success returns:

"P"	51H	30H	st0	st1	st2	Type	ATR

Failure returns:

"N"	51H	30H	e1	e0	ATR

C/D provides Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card, and once card is activated, it returns ATR.
Vcc = 30H: 5V to CPU card. Reset/Activate CPU card with EMV mode.
Vcc = 33H: 5V to CPU card. Reset/Activate CPU card with ISO7816 mode.
Vcc = 35H: 3V to CPU card. Reset/Activate CPU card with ISO7816 mode.
Vcc is an optional parameter; if command doesn't has this parameter, Vcc defaults to 30H.

In the process of RESET, if CPU card doesn't conform to EMV mode, e1 e0 ="69" error will be returned.
In the process of RESET, if testing IC card power fails, e1,e0 ="60" error will be returned.

Type: CPU card protocol type
= 30H T=0 protocol CPU card
= 31H T=1 protocol CPU card

ATR: Reset acknowledgement information format

TS	TO	TA1	TB1	……	TCK

6.1.2 CPU card power-off

HOST command:

"C"	51H	31H

Success returns:

"P"	51H	31H	st0	st1	st2

Failure returns:

"N"	51H	31H	e1	e0

Operate CPU card power-off.

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6.1.3 Read CPU card status

HOST command:

"C"	51H	32H

Success returns:

"P"	51H	32H	st0	st1	st2	Sti

Failure returns:

"N"	51H	32H	e1	e0

Inquire current CPU card status and return Sti status:

Sti = 30H: CPU card inactivated.

= 31H: CPU card activated with operating frequency 3.57 MHz.

= 32H: CPU card activated with operating frequency 7.16 MHz.

If IC card power circuit failure is detected, e1,e0 ="60" error will be returned.

6.1.4 T=0 protocol CPU card APDU operation

HOST command:

"C"	51H	33H	C-APDU

Success returns:

"P"	51H	33H	st0	st1	st2	R-APDU

Failure returns:

"N"	51H	33H	e1	e0

Operate reset/activated T=0 protocol CPU card APDU data exchange.

C-APDU: HOST sends T=0 CPU card data package, min 4 bytes and max 261 bytes. Format is as following:

CLA	INS	P1	P2	LC	Data 1	……	Le

R-APDU: CPU card returns HOST data package, min 2 bytes and max 258 bytes. Format is as following:

Data 1	……	Data (n)	Sw1	Sw0

If testing IC card power fails in the process of operation, e1,e0 ="60" error will be returned.
If current CPU card is not T=0 protocol, e1,e0 ="62" error will be returned.
If CPU card has a timeout, C/D will power off CPU card and return e1,e0 ="63" error.
If the other protocol error exists, C/D will power off CPU card and return e1,e0 ="64" error.
If HOST doesn't execute CPU card reset/activate operation, e1,e0 ="65" error will be returned.

Please refer to ISO/IEC7816-3 for T=0 APDU format and specific C-APDU command execution is subject to COS command.

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6.1.5 T=1 protocol CPU card APDU operation

HOST command:

"C"	51H	34H	C-APDU

Success returns:

"P"	51H	34H	st0	st1	st2	R-APDU

Failure returns:

"N"	51H	34H	e1	e0

Operate reset/activated T=1 protocol CPU card APDU data exchange.

C/D executes T=1 protocol to operate C-APDU card and combines C-APDU sent from HOST into an I-block to CPU card, and again returns R-APDU package extracted from I-block to HOST. Process is as following:

A. Send C-APDU

C-APDU

CLA	INS	P1	P2	Lc	Data 1	…	Data (Lc)	Le

↓

I-block

NAD	PCB	LEN	CLA	INS	P1	P2	Lc	Data 1	…	Data (Lc)	Le	EDC
Block head			Block information									Block end

B. Receive R-APDU

I-block

Block head			Block information									Block end
NAD	PCB	LEN	CLA	INS	P1	P2	LC	Data 1	…	Data (Lc)	Le	EDC

↓

R-APDU

CLA	INS	P1	P2	Lc	Data 1	…	Data (Lc)	Le

If testing IC card power fails in the process of operation, e1,e0 ="60" error will be returned.
If current CPU card is not T=1 protocol, e1,e0 ="62" error will be returned.
If CPU card has a timeout, C/D will power off CPU card and return e1,e0 ="63" error.
If the other protocol error exists, C/D will power off CPU card and return e1,e0 ="64" error.
If HOST doesn't execute CPU card reset/activate operation, e1,e0 ="65" error will be returned.

Please refer to ISO/IEC7816-3 for T=1 APDU format and specific C-APDU command execution is

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subject to COS command.

6.1.6 CPU card warm reset

HOST command:

"C"	51H	38H

Success returns:

"P"	51H	38H	st0	st1	st2	Type	ATR

Failure returns:

"N"	51H	38H	e1	e0

Operate C/D reset under IC card activation status and gain ATR.

Type: CPU card protocol type

= 30H T=0 protocol CPU card

= 31H T=1 protocol CPU card

6.1.7 Auto select T=0/T=1 protocol CPU card APDU operation

HOST command:

"C"	51H	39H	C-APDU

Success returns:

"P"	51H	39H	st0	st1	st2	R-APDU

Failure returns:

"N"	51H	39H	e1	e0

C/D auto determines current CPU card T=0/T=1 protocol, selects corresponding protocol C-APDU operation and returns R-APDU.

If testing IC card power fails in the process of operation, e1,e0 ="60" error will be returned.
If current CPU card is not T=0/T=1 protocol, e1,e0 ="62" error will be returned.
If CPU card has a timeout, C/D will power off CPU card and return e1,e0 ="63" error.
If the other protocol error exists, C/D will power off CPU card and return e1,e0 ="64" error.
If HOST doesn't execute CPU card reset/activate operation, e1,e0 ="65" error will be returned.

6.2 Operate SAM Card

6.2.1 SAM Card Cold Reset

HOST command:

"C"	52H	30H	Vcc

Success returns:

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"P"	52H	30H	st0	st1	st2	Type	ATR

Failure returns:

"N"	52H	30H	e1	e0	ATR

C/D provides Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card, and once card is activated, it returns ATR.

Type: SAM card protocol type

= 30H T=0 protocol CPU card

= 31H T=1 protocol CPU card

ATR: Reset acknowledgement information format

TS	TO	TA1	TB1	……	TCK

Vcc = 30H: 5V to SAM card. Reset/Activate CPU card with EMV mode.

Vcc = 33H: 5V to SAM card. Reset/Activate CPU card with ISO7816 mode.

Vcc = 35H: 3V to SAM card. Reset/Activate CPU card with ISO7816 mode.

Vcc is an optional parameter; if command doesn't has this parameter, Vcc defaults to 30H.

In the process of RESET, if SAM card doesn't conform to EMV mode, e1 e0 ="69" error will be returned.

In the process of RESET, if testing IC CARD power fails, e1,e0 ="60" error will be returned.

6.2.2 SAM Card Power-off

HOST command:

"C"	52H	31H

Success returns:

"P"	52H	31H	st0	st1	st2

Failure returns:

"N"	52H	31H	e1	e0

Operate activated SAM card power-off.

6.2.3 Read SAM Card Status

HOST command:

"C"	52H	32H

Success returns:

"P"	52H	32H	st0	st1	st2	Sti	Stj

Failure returns:

"N"	52H	32H	e1	e0

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Inquire current SAM card status and return Sti,Stj status:

Sti = 30H: SAM card inactivated.
= 31H: SAM card activated with operating frequency 3.57 MHz.
= 32H: SAM card activated with operating frequency 7.16 MHz.

Stj = 30H: Current SAM socket is No. 1.
= 31H: Current SAM socket is No. 2(optional).
= 32H: Current SAM socket is No. 3(optional).
= 33H: Current SAM socket is No. 4(optional).
= 34H: Current SAM socket is No. 5(optional).

If SAM card power circuit failure is detected, e1,e0 =”60” error will be returned.

6.2.4 T=0 protocol SAM Card APDU Operation

HOST command:

“C”	52H	33H	C-APDU

Success returns:

“P”	52H	33H	st0	st1	st2	R-APDU

Failure returns:

“N”	52H	33H	e1	e0

Operate reset/activated T=0 protocol SAM card data exchange.

If testing IC card power fails in the process of operation, e1,e0 =”60” error will be returned.
If current SAM card is not T=0 protocol, e1,e0 =”62” error will be returned.
If SAM card has a timeout, C/D will power off SAM card and return e1,e0 =”63” error.
If the other protocol error exists, C/D will power off SAM card and return e1,e0 =”64” error.
If HOST doesn’t execute SAM card reset/activate operation, e1,e0 =”65” error will be returned.

Please refer to ISO/IEC7816-3 for T=0 APDU format and specific C-APDU command execution is subject to COS command.

6.2.5 T=1 protocol SAM card APDU data exchange

HOST command:

“C”	52H	34H	C-APDU

Success returns:

“P”	52H	34H	st0	st1	st2	R-APDU

Failure returns:

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“N”	52H	34H	e1	e0

Operate reset/activated T=1 protocol SAM card data exchange.

If testing IC card power fails in the process of operation, e1,e0 =”60” error will be returned.
If current SAM card is not T=1 protocol, e1,e0 =”62” error will be returned.
If SAM card has a timeout, C/D will power off SAM card and return e1,e0 =”63” error.
If the other protocol error exists, C/D will power off SAM card and return e1,e0 =”64” error.
If HOST doesn’t execute SAM card reset/activate operation, e1,e0 =”65” error will be returned.

Please refer to ISO/IEC7816-3 for T=1 APDU format and specific C-APDU command execution is subject to COS command.

6.2.6 SAM card warm reset

HOST command:

“C”	52H	38H

Success returns:

“P”	52H	38H	st0	st1	st2	Type	ATR

Failure returns:

“N”	52H	38H	e1	e0

Operate C/D reset under IC card activation status and gain ATR.
Type: SAM card protocol type
= 30H T=0 protocol SAM card
= 31H T=1 protocol SAM card

6.2.7 Auto select T=0/T=1 protocol SAM card APDU operation

HOST command:

“C”	52H	39H	C-APDU

Success returns:

“P”	52H	39H	st0	st1	st2	R-APDU

Failure returns:

“N”	52H	39H	e1	e0

C/D auto determines current SAM card T=0/T=1 protocol, selects corresponding protocol C=APDU operation and returns R-APDU.

If testing IC card power fails in the process of operation, e1,e0 =”60” error will be returned.
If current SAM card is not T=0/T=1 protocol, e1,e0 =”62” error will be returned.

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If SAM card has a timeout, C/D will power off SAM card and return e1,e0 =”63” error.
If the other protocol error exists, C/D will power off SAM card and return e1,e0 =”64” error.
If HOST doesn’t execute SAM card reset/activate operation, e1,e0 =”65” error will be returned.

6.2.8 Select SAM socket

HOST command:

“C”	52H	40H	SAMn

Success returns:

“P”	52H	40H	st0	st1	st2

Failure returns:

“N”	52H	40H	e1	e0

Select SAM card on the SAM board.

SAMn = 30H: Select SAM 1.
= 31H: Select SAM 2(optional).
= 32H: Select SAM 3(optional).
= 33H: Select SAM 4(optional).
= 34H: Select SAM 5(optional).

This command is valid only for the C/D with PSAM board and can operate one SAM each time. SAM is selected automatically when C/D initializes.

6.3 Operate SLE4442 /4428 card

6.3.1 Reset/ Activate SLE4442/4428 card

HOST command:

“C”	53H	30H

Success returns:

“P”	53H	30H	st0	st1	st2	ATR(4 bytes)

Failure returns:

“N”	53H	30H	e1	e0

C/D provides Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card, and once card is activated, it returns ATR.

6.3.2 Power-off/ Deactivate SLE4442/4428 card

HOST command:

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“C”	53H	31H

Success returns:

“P”	53H	31H	st0	st1	st2

Failure returns:

“N”	53H	31H	e1	e0

C/D stops Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card, and card is power off and deactivated.

6.3.3 Read SLE4442/4428 card status

HOST command:

“C”	53H	32H

Success returns:

“P”	53H	32H	st0	st1	st2	Sti

Failure returns:

“N”	53H	32H	e1	e0

Inquire current SAM card status and return Sti status:

Sti = 30H: SLE4442/4428 card inactivated.
= 31H: SLE4442 card activated.
= 32H: SLE4428 card activated.

6.3.4 Operate SLE4442 card

All command data of SLE4442 (read/write) is operated in the way of ISO/IEC 7816 T=0 Standard data exchange command. Therefore, C/D does corresponding card operation when receiving specific command data. If command execution succeeds, add 9000H into normal return data package; if fails, return “sw1 + sw2” error code as normal return in the way of ISO/IEC 7816 T=0 Standard.

Sw1	Sw2	Descriptions
90H	00H	Operation succeeds.
6FH	00H	Operation fails.
6FH	01H	Key verifying fails.
6FH	02H	Key verifying fails and card is locked.
6BH	00H	Operation address overflows.
67H	00H	Operation length overflows.

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6.3.4.1Read SLE4442 main memory

HOST command:

“C”	53H	33H	00H	B0H	00H	abH	cdH

Success returns:

“P”	53H	33H	st0	st1	st2	Data

Failure returns:

“N”	53H	33H	e1	e0

ab: Read starting address of main memory
cd: Read data operating length

C/D operates SLE4442 main memory reading by specific abH and cdH parameters, and its main memory capacity is 256 bytes.

Read all memory data of SLE4442 card by following command:
Ex). “CR3”+00B0000000

6.3.4.2Read SLE4442 protection bits

HOST command:

“C”	53H	33H	00H	B0H	01H	abH	cdH

Success returns:

“P”	53H	33H	st0	st1	st2	Data

Failure returns:

“N”	53H	33H	e1	e0

ab: Read starting address of main memory
cd: Read data operating length

SLE4442 card 32bits’ protection bits status is presented by 4 bytes data, and corresponding protection bits’ address is 00H~1FH.

Read all protection bits data of SLE4442 card by following command:
Ex). “CR3”+00B0010004

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6.3.4.3Read SLE4442 security memory

HOST command:

"C"	53H	33H	00H	B0H	02H	abH	cdH

Success returns:

"P"	53H	33H	st0	st1	st2	Data

Failure returns:

"N"	53H	33H	e1	e0

ab: Read starting address of main memory
cd: Read data operating length

SLE4442 security memory is 4 bytes, respectively are: 1 byte key error counter data + 3 bytes key data values (key data could be read after correct key verifying).

Read all security memory data of SLE4442 card by following command:
Ex). "CR3"+00B0020004

6.3.4.4Write SLE4442 main memory

HOST command:

"C"	53H	33H	00H	D0H	00H	abH	cdH	efH…

Success returns:

"P"	53H	33H	st0	st1	st2	Data

Failure returns:

"N"	53H	33H	e1	e0

ab: Write starting address of main memory
cd: Write data operating length
ef: Data to be written (cdH byte)

Operate SLE4442 main memory data writing to write specific data to main memory and return operating result after C/D finishes data writing and verifying.
Correct verifying SLE4442 key is a must before data writing.
SLE4442 memory capacity is 256 bytes, and operating main memory writing could be done when cd = 00H.

Write all security memory data of SLE4442 card by following command:
Ex). "CR3"+00D000000+Write data (256 bytes)

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After command execution, return 9000H (operation success) or sw1,sw2 (operation failure). If specific data to be written is on data writing bit and writing protection status, data could not be written.

6.3.4.5Write SLE4442 data with protection bits

HOST command:

"C"	53H	33H	00H	D0H	01H	abH	cdH	efH…

Success returns:

"P"	53H	33H	st0	st1	st2	Data

Failure returns:

"N"	53H	33H	e1	e0

ab: Write starting address
cd: Write data operating length
ef: Data to be written (cdH byte)

Write memory unit with protection bits to main memory. Correct verifying SLE4442 key is a must before command execution.
SLE4442 protection bits writing is on main memory 00H~1FH. 00H~1FH memory units are subject to 32 bits writing protection status, e.g. bit0=1 of protection data byte0 is read, it means the memory unit of address 00H is with protection bits.
Once protection bits are set, writing protection bits cannot be changed.
E.g.: Write 20H to address 10H and operate writing protection bits at once.
Ex). "CR3"+00D001100120

After command execution, return 9000H (operation success) or sw1,sw2 (operation failure).
C/D will compare data of reading protection bits with data of protection bits to be written from receiving command, if it is different, C/D will not execute writing protection bits.
Writing protection bits could be operated one time only in main memory.

6.3.4.6Write SLE4442 security memory

HOST command:

"C"	53H	33H	00H	D0H	02H	abH	cdH	efH…

Success returns:

"P"	53H	33H	st0	st1	st2	Data

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Failure returns:

"N"	53H	33H	e1	e0

ab: Write starting address of main memory
cd: Write data operating length
ef: Data to be written (cdH byte)

After key verifying correct, 3bytes key in main memory is allowed to be changed.
E.g.: Change key (to 123456H) by following command:
Ex). "CR3"+00D0020103123456
After command execution, return 9000H (operation success) or sw1,sw2 (operation failure).
Notes: Pay attention to key error counter as it is allowed to be written, and counter is controlled by verifying key operation.

6.3.4.7Verify SLE4442 key

HOST command:

"C"	53H	33H	00H	20H	03H	01H	03H	efH...

Success returns:

"P"	53H	33H	st0	st1	st2	Data

Failure returns:

"N"	53H	33H	e1	e0

ef: Key data (3 bytes)

Verifying key is a must before changing SLE4442 data as it is card function requirement for operating next command.
Ex). "CR3"+0020030103xxxxxx (xxxxxx: 3bytes key data)
Card will compare its memory key with command data.
If user wants to change SLE4442 data, card key must to be known. Key error counter value will reset to zero from 2 or <2 and when it is zero, card will be locked and invalid.

6.3.5 Operate SLE4428 card

All command data of SLE4428 (read/write) is operated in the way of ISO/IEC 7816 T=0 Standard data exchange command. Therefore, C/D does corresponding card operation when receiving specific command data. If command execution succeeds, add 9000H into normal return data package; if fails, return "sw1 + sw2" error code as normal return in the way of ISO/IEC 7816 T=0 Standard.

Sw1	Sw2	Descriptions
90H	00H	Operation succeeds.

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6FH	00H	Operation fails.
6FH	01H	Key verifying fails.
6FH	02H	Key verifying fails and card is locked.
6BH	00H	Operation address overflows.
67H	00H	Operation length overflows.

6.3.5.1Read SLE4428 main memory

HOST command:

“C”	53H	34H	00H	B0H	0aH	bcH	deH

Success returns:

“P”	53H	34H	st0	st1	st2	Data

Failure returns:

“N”	53H	34H	e1	e0

abc: Read starting address of main memory

de: Read data operating length

C/D operates SLE4428 main memory reading by specific abcH and deH parameters, and its main memory capacity is 1024 bytes.

When de=00H, read 256 bytes data of SLE4428 card by following command:
Ex). “CR3”+00B0000000

6.3.5.2Read SLE4428 protection bits

HOST command:

“C”	53H	34H	00H	B0H	10H	abH	cdH

Success returns:

“P”	53H	34H	st0	st1	st2	Data

Failure returns:

“N”	53H	34H	e1	e0

ab: Read starting address of protection bits
cd: Read data operating length

Units with protection writing function in SLE4428 main memory are 1024bytes and corresponding writing protection data is 1024bits, which means reading protection bits are (1024=128x8). Protection bits data is composed of 1 byte from address 000H~007H.

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Read all protection bits data of SLE4428 card by following command:
Ex). "CR3"+00B0010004

C/D reads protection bits data of cdH length on the basis of abH parameter specific starting address.

6.3.5.3Write SLE4428 main memory

HOST command:

"C"	53H	34H	00H	D0H	0aH	bcH	deH	fgH…

Success returns:

"P"	53H	34H	st0	st1	st2	Data

Failure returns:

"N"	53H	34H	e1	e0

abc: Write starting address of main memory
de: Write data operating length
fg: Data to be written

Operate SLE4428 main memory data writing to write specific data to main memory and return operating result after C/D finishes data writing and verifying.
Correct verifying SLE4428 key is a must before data writing.
SLE4428 memory capacity is 1024 bytes.

Write memory data 256 bytes of SLE4428 card by following command:
Ex). "CR4"+00D000000+Write data (256 bytes)

After command execution, return 9000H (operation success) or sw1,sw2 (operation failure). If specific data to be written is on data writing bit and writing protection status, data could not be written.

6.3.5.4Write SLE4428 data with verifying

HOST command:

"C"	53H	34H	00H	D0H	1aH	bcH	deH	fgH…

Success returns:

"P"	53H	34H	st0	st1	st2	Data

Failure returns:

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“N”	53H	34H	e1	e0

abc: Write starting address of main memory
de: Write data operating length
fg: Data to be written (deH byte)

Operate SLE4428 main memory data writing to write specific data to main memory and return operating result after C/D finishes data writing and verifying.
Correct verifying SLE4428 key is a must before data writing.

6.3.5.5Write SLE4428 Data with protection bits

HOST command:

“C”	53H	34H	00H	D0H	2aH	bcH	deH	fgH…

Success returns:

“P”	53H	34H	st0	st1	st2	Data

Failure returns:

“N”	53H	34H	e1	e0

abc: Write starting address with protection bits
de: Write data operating length
fg: Data to be written (deH byte)

Write memory unit with protection bits to main memory. Correct verifying SLE4428 key is a must before command execution.

After command execution, return 9000H (operation success) or sw1,sw2 (operation failure).
C/D will compare data of reading protection bits with data of protection bits to be written from receiving command, if it is different, C/D will not execute writing protection bits.
Writing protection bits could be operated only when writing data is the same as memory data..

6.3.5.6Verify SLE4428 key

HOST command:

“C”	53H	34H	00H	20H	00H	00H	02H	efH…

Success returns:

“P”	53H	34H	st0	st1	st2	Data

Failure returns:

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“N”	53H	34H	e1	e0

ef: Key data (2 bytes)

Verifying key is a must before changing SLE4428 data as it is card function requirement for operating next command.
Ex). “CR3”+0020000002xxxx (xxxx: 2bytes key data)
Card will compare its memory key with command data.
If user wants to change SLE4428 data, card key must to be known. Key error counter value will reset to zero from 7 or <7 and when it is zero, card will be locked and invalid.

6.4 Operate I2C memory card

6.4.1 Reset/Activate I2C card

HOST command:

“C”	54H	30H	Wrd	Vcc

Success returns:

“P”	54H	30H	st0	st1	st2

Failure returns:

“N”	54H	30H	e1	e0

Operate I2C (24C01, 24C02, 24C04, 24C08, 24C16, 24C32, 24C64, 24C128, 24C256) card reset/activate.
C/D provides Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card.
Wrd sets I2C card type:

Wrd= 30H Auto activate (24C01, 24C02, 24C04, 24C08, 24C16, 24C32, 24C64, 24C128, 24C256) card
Wrd = 31H Activate 24C01 card
Wrd = 32H Activate 24C02 card
Wrd = 33H Activate 24C04 card
Wrd = 34H Activate 24C08 card
Wrd = 35H Activate 24C16 card
Wrd = 36H Activate 24C32 card
Wrd = 37H Activate 24C64 card
Wrd = 38H Activate 24C128 card
Wrd = 39H Activate 24C256 card

Vcc selects voltage for card:

Vcc = 30H C/D provides 5V to reset/activate card.
Vcc = 31H C/D provides 3V to reset/activate card.

Vcc is an optional parameter; if command doesn’t has Set parameter, Vcc defaults to Set=30H.

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6.4.2 Power-off/Deactivate I2C card

HOST command:

"C"	54H	31H

Success returns:

"P"	54H	31H	st0	st1	st2

Failure returns:

"N"	54H	31H	e1	e0

C/D stops Vcc (power supply), CLK (clock signal) and RST (reset signal) to dispense card.

6.4.3 Read I2C card status

HOST command:

"C"	54H	32H

Success returns:

"P"	54H	32H	st0	st1	st2	Sti

Failure returns:

"N"	54H	32H	e1	e0

Read status and return Sti status for correct execution:

Sti status meaning:

Sti = 30H: No I2C card activated.
Sti = 31H Activate 24C01 card
Sti = 32H Activate 24C02 card
Sti = 33H Activate 24C04 card
Sti = 34H Activate 24C08 card
Sti = 35H Activate 24C16 card
Sti = 36H Activate 24C32 card
Sti = 37H Activate 24C64 card
Sti = 38H Activate 24C128 card
Sti = 39H Activate 24C256 card

6.4.4 Read/Write I2C card

All command data of I2C (read/write) is operated in the way of ISO/IEC 7816 T=0 Standard data (C-APDU) format exchange command. Therefore, C/D does corresponding card operation when receiving specific command data. If command execution succeeds, add 9000H into normal return data package; if fails, return "sw1 + sw2" error code as normal return in the way of ISO/IEC 7816-3 T=0 Standard.

Sw1	Sw2	Descriptions

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90H	00H	Operation succeeds.
6FH	00H	Operation fails.
6BH	00H	Operation address overflows.
67H	00H	Operation length overflows.

Please note corresponding operation address range when r/w I2C card:

Card_type	ab,cd
24C01	0000H ~ 007FH
24C02	0000H ~ 00FFH
24C04	0000H ~ 01FFH
24C08	0000H ~ 03FFH
24C16	0000H ~ 07FFH
24C32	0000H ~ 0FFFH
24C64	0000H ~ 1FFFH
24C128	0000H ~ 3FFFH
24C256	0000H ~ 7FFFH

6.4.5 Read I2C card

HOST command:

“C”	54H	33H	00H	B0H	abH	cdH	efH…

Success returns:

“P”	54H	33H	st0	st1	st2	Data

Failure returns:

“N”	54H	33H	e1	e0

ab: Read operating high byte address
cd: Read operating low byte address
ef: Read operation length

C/D writes efH specific length I2C card data on basis of abH,cdH specific starting address. efH specific operating length cannot exceed I2C card address range.
Read 8 bytes data from I2C card by following command:
Ex). “CU3”+00B0000008

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6.4.6 Write I2C card

HOST command:

"C"	54H	34H	00H	D0H	abH	cdH	efH	ghH…

Success returns:

"P"	54H	34H	st0	st1	st2	Data

Failure returns:

"N"	54H	34H	e1	e0

ab: Write operating high byte address
cd: Write operating low byte address
ef: Write operation length
gh: Write data (efH byte)

C/D reads efH specific length I2C card data on basis of abH,cdH specific starting address. efH specific operating length cannot exceed I2C card address range.
Write 8 bytes data from I2C card by following command:
Ex). "CU3"+00B0000008+ write data (8 bytes)
After command execution, return 9000H (operation success) or sw1,sw2 (operation failure).

7 Contactless IC Card Operate

7.1 Operate RF card (contactless IC card)

7.1.1 Reset/Activate RF card

HOST command:

"C"	60H	30H	Set1	Set2

(1) Success returns of Mifare one card:

"P"	60H	30H	st0	st1	st2	Rtype	ATQA	UID_len	UID_data	SAK

Failure returns of Mifare one card:

"N"	60H	30H	e1	e0	Rtype	ATQA	UID_len	UID_data	SAK

(2) Success returns of ISO14443 Type A card:

"P"	60H	30H	st0	st1	st2	Rtype	ATQA	UID_len	UID_data	SAK	ATS

Failure returns of ISO14443 Type A card:

"N"	60H	30H	e1	e0	Rtype	ATQA	UID_len	UID_data	SAK	ATS

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(3) Success returns of ISO14443 Type B card:

"P"	60H	30H	st0	st1	st2	Rtype	ATQB

Failure returns of ISO14443 Type B card:

"N"	60H	30H	e1	e0	Rtype	ATQB

Operate reset/activate RF card.
C/D supports IEC/ISO14443 Type A and Type B card reset/activate.

C/D manages RF card reset/activate as following process:

1) Mifare one card
1. Request A(REQ A) / Answer Request A (ATQ A)
2. Anticollision
3. Select(SEL) /Unique Identifier(UID) & Select Acknowledge(SAK)

When Mifare one card is activated successfully, C/D returns:
Answer to Request of Type A (ATQA, 2 bytes)
Unique Identifier (UID_data, 4~10 bytes)
Select Acknowledge (SAK, 1 byte).

2)ISO14443Type A
1. Request A(REQ A) /Answer Request A (ATQ A).
2. Anticollision
3. Select(SEL) /Unique Identifier(UID) & Select Acknowledge(SAK)
4. Request for answer to select (RATS) /Answer to Select(ATS)
5. Protocol and parameter selection request(PPSR) /PPS start(PPSS)

When ISO/IEC 14443 Type A card is activated successfully, C/D returns:
Answer to Select (ATS, 1 byte)
Protocol and Parameter Selection Request (PPSR, 1 byte)

3)ISO14443 Type B
1. Request B(REQ B) /Answer Request B (ATQ B)
2. Attribute(ATTRIB) /Answer to ATTRIB

When ISO/IEC 14443 Type B card is activated successfully, C/D returns ATQB (12 bytes) which includes:
50H, PUPI (4 bytes), App data (4 byte), Protocol info (3 bytes)
Set1,Set2 set reset/activate sequence of RF card with different protocol. Effective values are:
41H ('A'= Type A), 42H ('B'= Type B)), 30H ('0'= not used)

Ex1: Set1='A', Set2 ='B' (default)
It means reset/activate sequence is: Type A protocol is the first reset/activate sequence, Type B protocol is the second reset/activate sequence.

Ex2: Set1='B', Set2 ='A'
It means reset/activate sequence is: Type B protocol is the first reset/activate sequence, Type A protocol is the second reset/activate sequence.

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Ex3: Set1='A', Set2 ='0'
It means reset/activate sequence is: Type A protocol is the first reset/activate sequence, Type B protocol is deactivated.

Ex4: Set1='B', Set2 ='0',
It means reset/activate sequence is: Type B protocol is the first reset/activate sequence, Type A protocol is deactivated.

Rtype: current protocol for activating card
= 41H ('A') Current card conforms to ISO/IEC 14443 Type A protocol.
= 42H ('B') Current card conforms to ISO/IEC 14443 Type B protocol.
= 4DH ('M') Current card conforms to Philps Mifare one card protocol.

When Rtype=4DH ('M')
ATQA= 0044H is Mifare Ultralight Card.
ATQA= 0004H is Mifare S50 1K Card.
ATQA= 0002H is Mifare S70 4K Card.

Mifare one, ISO/IEC 14443 Type A card returns UID_len specific UID_data information length.
UID_len=4 means returned UID_data information length is 4 bytes.
UID_len=7 means returned UID_data information length is 7 bytes.
UID_len=10 means returned UID_data information length is 10 bytes.

7.1.2 Power-off/Deactivate RF card

HOST command:

"C"	60H	31H

Success returns:

"P"	60H	31H	st0	st1	st2

Failure returns:

"N"	60H	31H	e1	e0

Operate RF card power-off/deactivate and RF module chip closes all antenna output signal.

7.1.3 Inquire RF card status

HOST command:

"C"	60H	32H

Success returns:

"P"	60H	32H	st0	st1	st2	sti	stj

Failure returns:

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“N”	60H	32H	e1	e0

Inquire current RF card status sti,stj:

sti	stj	Descriptions
‘0’	‘0’	No RF card activated.
‘1’	‘0’	Mifare one S50 card.
	‘1’	Mifare one S70 card.
	‘2’	Mifare one UL card.
‘2’	‘0’	Type A CPU card.
‘3’	‘0’	Type B CPU card.

7.1.4 Operate Mifare 1 card

All command data of Mifare 1 card (read/write) is operated in the way of ISO/IEC 7816 T=0 Standard data (C-APDU) format exchange command. Therefore, C/D does corresponding card operation when receiving specific command data. If command execution succeeds, add 9000H into normal return data package; if fails, return “sw1 + sw2” error code as normal return in the way of ISO/IEC 7816-3 T=0 Standard.

Sw1	Sw2	Descriptions
90H	00H	Operation succeeds.
6FH	00H	Operation fails.
6BH	00H	Operation address overflows.
67H	00H	Operation length overflows.

7.1.4.1Verify key

HOST command:

“C”	60H	33H	00H	20H	ks	sn	lc	pdata

Success returns:

“P”	60H	33H	st0	st1	st2	rdata

Failure returns:

“N”	60H	33H	e1	e0

Download keys to C/D to directly verify sector keys:

ks (1 byte): Key select (Key A=00H, Key B=01H).
sn (1 byte): Sector number (S50 card sn=00H~0FH, S70 card sn=00H~27H).
lc (1 byte): Length of key data (lc=06H).
pdata (6 bytes): Password data.
rdata (2 bytes): Return data. Success returns 9000, failure returns sw1 + sw2 (2 bytes).

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7.1.4.2 Load key verifying from EEPROM

HOST command:

"C"	60H	33H	00H	21H	ks	sn

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Read stored key from EEPROM of C/D RF module to verify sector keys.
Pre-load keys to EEPROM by 9.10.4.4 command.
EEPROM of C/D RF module can store 32 sets of key data.
ks (1 byte): Key select (Key A=00H, Key B=01H).
sn (1 byte): Sector number (sn=00H~0FH).
rdata (2 bytes): Return data. Success returns 9000.

7.1.4.3 Modify Sector Key (KEY A)

HOST command:

"C"	60H	33H	00H	D5H	00H	sn	lc	pdata

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Operate sector key (KEY A) modification.
This command execution can only do modification to KEY A and rewrite KEY B to "0xFF, 0xFF, 0xFF, 0xFF, 0xFF, 0xFF", meanwhile, write control word to "0xFF, 0x07, 0x80, 0x69" (factory default).
If KEY A, KEY B and control word need to be modified, use block writing command to execute.
sn (1 byte): Sector number (S50 card sn=00H~0FH, S70 card sn=00H~27H).
lc (1 byte): Length of key data (lc=06H).
pdata (6 bytes): New password data.
rdata (2 bytes): Return data. Success returns 9000, failure returns sw1 + sw2 (2 bytes).

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7.1.4.4Download Keys to EEPROM

HOST command:

“C”	60H	33H	00H	D0H	ks	sn	lc	pdata

Success returns:

“P”	60H	33H	st0	st1	st2	rdata

Failure returns:

“N”	60H	33H	e1	e0

Download keys to EEPROM of C/D RF module to directly verify sector keys:
EEPROM of C/D RF module can store 32 sets of key data:
ks (1 byte): Key select (Key A=00H, Key B=01H).
sn (1 byte): Sector number (sn=00H~0FH).
lc (1 byte): Length of key data (lc=06H).
pdata (6 bytes): Password data.
rdata (2 bytes): Return data.
Success returns sw1 + sw2 = 9000H, failure returns sw1 + sw2 = 6F00H.

7.1.4.5Read sector block data

HOST command:

“C”	60H	33H	00H	B0H	sn	bn	le

Success returns:

“P”	60H	33H	st0	st1	st2	rdata

Failure returns:

“N”	60H	33H	e1	e0

Read one block data or continuously read multi blocks data from RF sector.
sn (1 byte): Sector number.
bn (1 byte): Starting block number.
le (1 byte): Block length
(le=01H read one block data, le=03H continuously read three blocks data.).
rdata (2 bytes): Return data. Success returns +9000H, failure returns sw1 + sw2 (2 bytes).

Notes:
1. Ultralight card’s each sector has one block only, and each block only has 4 bytes data. Each block of S50, S70 card has 16 bytes data.
2. Sector number, starting block number and block length’s value range of Ultralight Card, Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range.
Ultralight Card: sn=00H-0FH, bn=00H, le=01H-0FH

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Mifare 1k(S50):	sn=00H-0FH,	bn=00H-03H,	le=01H-04H
Mifare 1k (S70):	sn=00H-20H,	bn=00H-03H,	le=01H-04H
	sn=21H-27H,	bn=00H-0FH,	le=01H-10H

(Last 8 sectors of S70 card have 16 blocks of each sector.)

7.1.4.6Write Sector Block Data

HOST command:

"C"	60H	33H	00H	D1H	sn	bn	lc	wdata

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Write one block data or continuously read multi blocks data of RF sector.

sn (1 byte):	Sector number.
bn (1 byte):	Starting block number.
lc (1 byte):	Block length.
wdata:	Block data to be written (n bytes).
rdata (2 bytes):	Return data.
	Success returns block data read +9000H, failure returns sw1+sw2(2 bytes).

Notes:

1. Ultralight Card's each sector has one block only, and each block only has 4 bytes data. Each block of S50, S70 card has 16ytes data.
2. Sector number, starting block number and block length's value range of Ultralight Card, Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range.

Ultralight Card:	sn=00H-0FH,	bn=00H-03H,	lc =01H-03H
Mifare 1k(S50):	sn=00H-0FH,	bn=00H-03H,	lc =01H-03H
Mifare 1k (S70):	sn=00H-20H,	bn=00H-03H,	lc =01H-03H
	sn=21H-27H,	bn=00H-0FH,	lc =01H-0FH

(Last 8 sectors of S70 card have 16 blocks of each sector.)

3. Last block of S50, S70 card's each sector is the control block of that sector to store KEY A, read/write control word and KEY B. Please do not miswrite this block when doing multi blocks writing and C/D is not allowed to operate sector's last block when continuously doing multi blocks data writing.

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7.1.4.7Initialization value

HOST command:

"C"	60H	33H	00H	D2H	sn	bn	lc	wdata

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Operate initialization value of RF card sector's block.

sn (1 byte): Sector number.
bn (1 byte): Starting block number.
lc (1 byte): Block length of initialization value (lc=04H).
wdata: Initialization value data (4 bytes).
rdata (2 bytes): Return data.
Success returns 9000H, failure only returns sw1 + sw2 (2 bytes).

Notes: Sector number and starting block number of Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range and please do not do value initialization of each sector's last block.

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k (S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,
(Last 8 sectors of S70 card have 16 blocks of each sector.)

7.1.4.8Read Value

HOST command:

"C"	60H	33H	00H	B1H	sn	bn

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Operate value reading of RF card sector block.

sn (1 byte): Sector number
bn (1 byte): Starting block number
rdata: Return data. Success returns value read (4 bytes) +9000H, failure only returns sw1 + sw2 (2 bytes).

Notes: Sector number and starting block number of Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range and please do not do value initialization of each sector's last block.

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Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k (S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,
(Last 8 sectors of S70 card have 16 blocks of each sector.)

7.1.4.9Increment

HOST command:

"C"	60H	33H	00H	D3H	sn	bn	lc	wdata

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Operate increment of RF card sector block.
sn (1 byte): Sector number
bn (1 byte): Starting block number
lc (1 byte): Increment data length (lc=04H)
wdata: Increment data (4 bytes)
rdata: Return data.
Success returns 9000H, failure only returns sw1 + sw2 (2 bytes).

Notes: Sector number and starting block number of Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range and please do not do value initialization of each sector's last block.
Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k (S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,
(Last 8 sectors of S70 card have 16 blocks of each sector.)

7.1.4.10 Decrement

HOST command:

"C"	60H	33H	00H	D4H	sn	bn	lc	wdata

Success returns:

"P"	60H	33H	st0	st1	st2	rdata

Failure returns:

"N"	60H	33H	e1	e0

Operate decrement of RF card sector block.

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sn (1 byte): Sector number
bn (1 byte): Starting block number
lc (1 byte): Decrement data length (lc=04H)
wdata: Decrement data (4 bytes)
rdata: Return data.
Success returns 9000H, failure only returns sw1 + sw2 (2 bytes).

Notes: Sector number and starting block number of Mifare 1k(S50), Mifare 1k (S70) card cannot exceed card capacity range and please do not do value initialization of each sector's last block.

Mifare 1k(S50): sn=00H-0FH, bn=00H-03H,
Mifare 1k (S70): sn=00H-20H, bn=00H-03H,
sn=20H-27H, bn=00H-0EH,
(Last 8 sectors of S70 card have 16 blocks of each sector.)

7.1.5 Type A RF card communication

HOST command:

"C"	60H	34H	C-APDU

Success returns:

"P"	60H	34H	st0	st1	st2	R-APDU

Failure returns:

"N"	60H	34H	e1	e0

This command operates card data exchange of RF Type A T=CL protocol according to ISO/IEC 14443-4 specification.
Notes: Maximum length of C-APDU package is 261 bytes, R-APDU package 258 bytes.

7.1.6 Type B RF card communication

HOST command:

"C"	60H	35H	C-APDU

Success returns:

"P"	60H	35H	st0	st1	st2	R-APDU

Failure returns:

"N"	60H	35H	e1	e0

This command operates card data exchange of RF Type B T=CL protocol according to ISO/IEC 14443-4 specification.
Notes: Maximum length of C-APDU package is 261 bytes, R-APDU package 258 bytes.

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8 Extend Command Descriptions

8.1 Read serial number

HOST command:

"C"	A2H	30H

Success returns:

"P"	A2H	30H	st0	st1	st2	len	ICRW_SN

Failure returns:

"N"	A2H	30H	e1	e0

Operate RF card wakeup/sleep.

Set=30H Operate RF card sleep.

Set=31H Operate RF card wakeup.

len: Read C/D S/N data length (min. 0 byte; max. 18 bytes)

ICRW_SN: C/D serial number.

8.2 Read configuration information

HOST command:

"C"	A3H	30H

Success returns:

"P"	A3H	30H	st0	st1	st2	ICRW_Config

Failure returns:

"N"	A3H	30H	e1	e0

Descriptions of reading C/D configuration information (S1~S10 are 10 bytes in total):

NAME	VALUE	DESCRIPTIONS	DETAILS
S1		SK dispenser type option	SYNCOTEK dispenser identifier.
	"7"	S1="7"	SK-AD3 identifier word S1=37H.
S2/S3/S7 (3 bytes)		User Code option	User software code.
	"V10"	SYNCOTEK version	SYNCOTEK standard program.
	"XXX"		Client customized version.
S5		Card r/w type option	Card type supported.
	"0"		With dispensing but without r/w function.
	"I"	IC card r/w	Supports contact IC card r/w only.
	"C"	RF card r/w	Supports RF card r/w only.
	"E"	IC + RF card r/w	Supports IC + RF card r/w.
S6		Interface type option	Communication interface type.
	"R"	RS-232 interface type	RS232 communication interface.
S7		IC card write type	IC card r/w function type.
	"0"		No IC card r/w.

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	“1”		With IC contact for 3rd-party module.
	“2:”		SYNCOTEK standard IC card r/w.
S8		RF card write type	RF card r/w function type.
	“0”		No RF card r/w.
	“1”		With RF antenna for 3rd-party module.
	“2:”		SYNCOTEK standard RF card r/w.
“S9”		SAM option	
	“0”	Not SAM	No SAM card function.
	“1”	SAM 1	Supports one SAM card.
	“2:”	SAM 2	Supports two SAM cards.
	“3”	SAM 3	Supports three SAM cards.
	“4”	SAM 4	Supports four SAM cards.
	“5”	SAM 5	Supports five SAM cards.
S10			Dispensing mode.
	“0”		Dispensing mode with rubbing wheel.
	“1”		Dispensing mode with hook.

8.3 Read version information

HOST command:

“C”	A4H	Pm

Success returns:

“P”	A4H	30H	st0	st1	st2	Rev.

Failure returns:

“N”	A4H	30H	e1	e0

Read C/D software relevant version information.

Pm=30H Read C/D software version information.

Ex: Rev=

Pm=30H Read IC card software version information.

Ex: Rev =“ICCARD_V10_A_090910”

Pm=30H Read RF card software version information.

Ex: Rev =“RFCARD_V10_A_090910”

8.4 Counter function

8.4.1 Read counter value

HOST command:

Room 716, Yiben E-Commerce
and Industrial Park, Chaguang Road
Nanshan District, Shenzhen, China
Post Code: 518055
Tel: +86-755-82706901
Fax: +86-755-82706900
Email: sales@syncotek.com
Http://www.syncotek.com

“C”	A5H	30H

Success returns:

“P”	A5H	30H	st0	st1	st2	Count (3 bytes)

Failure returns:

“N”	A5H	30H	e1	e0

Read counter value, each recycling card adds 1 to counter value when starting counter function in reset command.

Count= “000” ~ “999”: Counter value, min. 0 and max. 999. When counter value exceeds 999, report counter overflow error (e1,e0=”50”).

8.4.2 Set counter initial value

HOST command:

“C”	A5H	31H	Count (3 bytes)

Success returns:

“P”	A5H	31H	st0	st1	st2

Failure returns:

“N”	A5H	31H	e1	e0

Count= “000” ~ “999”: Counter value, min. 0 and max. 999